

Question	Marking Guidance	Mark	Comments
05.1	1. (Colour vision involves) <u>cones</u>; 2. (Each type of) photoreceptor has a different pigment OR (Each type of) photoreceptor absorbs particular/different (range of) wavelength(s) OR (Each type of) photoreceptor stimulated by particular/different (range of) wavelength(s); 3. Greater absorption by 'red sensitive' than 'green sensitive' (cells/photoreceptors/cones) OR Provides percentage values which indicate difference in light absorption (at 600nm) OR More impulses to brain from 'red sensitive' than 'green sensitive' (photoreceptors/cones) OR More impulses along optic nerve from 'red sensitive' than 'green sensitive' (photoreceptors/cones);	3 (3 x AO2)	1, 2 and 3. Reject 'red cones/photoreceptors' and 'green cones/photoreceptors' only once. 1 and 2. Each cone has a different pigment or absorbs particular wavelengths = two marks. 1 and 3. Greater absorption by 'red sensitive' than 'green sensitive' <u>cones</u> = two marks. 3. Allow approximately correct percentage values. 3. Accept suitable alternatives for 'sensitive' e.g. detecting/absorbing. 3. Accept action potentials for impulses.

			3. Do not credit 'signals', 'messages' for third or fourth options for mark point 3.
--	--	--	---

Question	Marking Guidance	Mark	Comments
05.2	Box 3 correct - Several photoreceptors connecting to one neurone and spatial summation	1 (AO1)	

Question	Marking Guidance	Mark	Comments
05.3	1. Geographical isolation/separation due to elevation/altitude; 2. Allopatric speciation due to isolation/separation; 3. Different selection pressures OR Different environment(s); 4. (However) some overlap in distribution OR (Both) plants found in same area/habitat/altitude; 5. So (possibly) sympatric speciation; 6. (Variation due to) mutation(s); 7. Reproductive isolation/separation due to different pollinators/distributions/altitudes OR Separate gene pools due to different pollinators/distributions/altitudes; 8. Change in allele frequency (in each population) OR Different allele frequency (in each population); 9. Different species) can no longer (interbreed to) produce fertile offspring;	5 max (5 x AO3)	3. Accept selection (due to) flower colour. 3. Accept different 'abiotic conditions/factors' for 'different environment(s)' or different named factor e.g. temperature, humidity but 'different altitudes' on its own is not enough. 5. Only awarded if mark point 4 is credited. 6. Reject mutation(s) if context incorrect e.g. 'mutate to adapt'. 7. Accept 'no gene flow' for separate gene pools. 8. Accept 'increase' or 'decrease' for 'change'.
